

changes direction. (Using numbers also makes sense because moving right involves adding to each alien's x -coordinate value, and moving left involves subtracting from each alien's x -coordinate value.)

Checking Whether an Alien Has Hit the Edge

We need a method to check whether an alien is at either edge, and we need to modify `update()` to allow each alien to move in the appropriate direction. This code is part of the Alien class:

```
alien.py    def check_edges(self):
            """Return True if alien is at edge of screen."""
            screen_rect = self.screen.get_rect()
            ❶    return (self.rect.right >= screen_rect.right) or (self.rect.left <= 0)

            def update(self):
                """Move the alien right or left."""
                ❷    self.x += self.settings.alien_speed * self.settings.fleet_direction
                self.rect.x = self.x
```

We can call the new method `check_edges()` on any alien to see whether it's at the left or right edge. The alien is at the right edge if the right attribute of its `rect` is greater than or equal to the right attribute of the screen's `rect`. It's at the left edge if its left value is less than or equal to 0 ❶. Rather than put this conditional test in an `if` block, we put the test directly in the `return` statement. This method will return `True` if the alien is at the right or left edge, and `False` if it is not at either edge.

We modify the method `update()` to allow motion to the left or right by multiplying the alien's speed by the value of `fleet_direction` ❷. If `fleet_direction` is 1, the value of `alien_speed` will be added to the alien's current position, moving the alien to the right; if `fleet_direction` is -1, the value will be subtracted from the alien's position, moving the alien to the left.

Dropping the Fleet and Changing Direction

When an alien reaches the edge, the entire fleet needs to drop down and change direction. Therefore, we need to add some code to `AlienInvasion` because that's where we'll check whether any aliens are at the left or right edge. We'll make this happen by writing the methods `_check_fleet_edges()` and `_change_fleet_direction()`, and then modifying `_update_aliens()`. I'll put these new methods after `_create_alien()`, but again, the placement of these methods in the class isn't critical.

```
alien
_invasion.py    def _check_fleet_edges(self):
                """Respond appropriately if any aliens have reached an edge."""
                ❶    for alien in self.aliens.sprites():
                    if alien.check_edges():
                        ❷    self._change_fleet_direction()
                        break

                def _change_fleet_direction(self):
```
